

Seeking assignments in Metallurgical Operations with an organisation of repute.

AN OVERVIEW

- ☞ A competent professional with over 8 years of experience in Failure analysis, Corrosion Testing Heat treatment, Metallurgical lab setup, Testing & Inspection, Research and Development.
- ☞ ***Presently working with Lone star Alpha Laboratories, Muscat, Oman*** as a Senior Metallurgist and Corrosion Engineer.

TECHNICAL CAPABILITY

Metallurgical Operations

- ☞ As a Metallurgist and Corrosion Engineer in metallurgical testing laboratories.
- ☞ Undertaking metallographic analysis of plain carbon steel, alloy steel, tool steel, and stainless steel, non ferrous steel like copper, titanium and super alloy.
- ☞ Developing materials and process for new applications.
- ☞ To Carry out Corrosion testing and evaluation as per respective ASTM and NACE standard.

Testing / Inspection

- ☞ Metallurgical analysis of steels and stainless steel.
- ☞ Evaluating technical qualification metals.
- ☞ Inspecting & maintaining quality standards for raw materials, ensuring stringent adherence to standards.

KNOWLEDGE OUTLINE

Metallurgical

- ☞ Metallographic and mechanical testing for metals and alloys for properties like hardness, tensile strength, Compression strength, 3 point bend test, impact test as per ASTM and IS standard.
- ☞ Knowledge about Duplex and super duplex stainless steel material testing and evaluation.
- ☞ To carry out In-situ Metallography testing on site and to conduct the evaluation of microstructure.

FAILURE ANALYSIS

- ☞ Complete knowledge identified and suggested corrective action plans for actual and potential causes of failure in root cause analysis and advanced metrology tools (SEM, EDS and XRD.)
- ☞ Provided Root Cause Corrective Action support for numerous continuous product improvement projects with a focus on steel and iron castings, forgings, and heat treatment.
- ☞ Coordinates failure analysis with other disciplines as necessary to lead to determination of root cause.
- ☞ Complete knowledge Writing the technical reports metallurgical results.

Heat treatment

- ☞ Complete knowledge of Vacuum heat treatment of stainless steel.
- ☞ Complete knowledge of sealed quench furnace (SQF) of steels.
- ☞ Heat treatment process design for induction hardening and tempering, Full hardening and tempering.
- ☞ Handled process like carburizing, carbonitriding, nitriding, nitro carburizing.

LEARNING ACCOUNT

Year	Name of degree	Name of institute	Percentage
2013-2016	B.E(Metallurgical Engineering)	Government College of Engineering, Salem.	74.41
2011-2013	Diploma	The Kaveri polytechnic college Mecheri-11	90.67
2011	HSC	Govt. higher sec school, Vanavasi	57
2009	State Board	Vaidheeswara higher sec school Mettur dam	60.50

OCCUPATIONAL OUTLINE

Since Nov'20 with Lonestar Alpha Laboratories as Senior Metallurgical Engineer, in Metallurgical Lab.

Role:

- ☞ Carry out Metallurgical examination to understand the inherent microstructure as well as defects or abnormalities present in the material. It includes the determination of ASTM grain size, Degree of banding, Nitriding depth, Case hardening depth, Inclusion count, manual count for ferrite content etc.
- ☞ Carrying out Ferrite Content Measurement at sites using FERRITESCOPE and Manual count method as per ASTM E 562.
- ☞ Carry out In-situ Replica Metallographic on Tubes, Pipes, shell plates of tanks in Gas Refineries and on Engineering components and equipments to find out the micro structural changes and to assess metallurgical conditions of the equipment. Prepare detailed reports with Micrographs, Microstructure Description, and Discussion with Interpretation, Results and Conclusion.
- ☞ Knowledge about all standard which is related to Mechanical and Metallurgical testing like ASTM,ISO,BS,NACE,API,ASME,PDO Standard(All SP version) etc..
- ☞ knowledge about PQR,WQT testing as per PDO and related other standard which is suitable for mechanical and metallurgical testing.

Corrosion Test Knowledge:

- ☞ NACE TM-0284-Hydrogen Induced Cracking.
- ☞ NACE TM-0177-Four point bend Sulfide stress Corrosion cracking Test.
- ☞ ASTM G 48 Method A & C- Pitting and Crevice Corrosion Resistance of Stainless Steels and Related Alloys by Use of Ferric Chloride Solution.
- ☞ ASTM G 28 - Detecting Susceptibility to Intergranular Attack in Wrought Nickel-Rich Chromium Bearing Alloys .
- ☞ ASTM 262 Practice A,B,C & E - Detecting Susceptibility to Intergranular Attack in Austenitic Stainless steel.
- ☞ ASTM A 923 Method A & C - Detecting Detrimental Intermetallic Phase in Duplex Austenitic/Ferritic Stainless Steel.

June'17 - Oct' 20 with Tii techno Testing services pvt ltd -Metallurgist.

Tii Techno Testing Services is A state of the art testing and calibration facilities in Chennai, India (ISO/IEC 17025 - NABL Accredited Testing and Calibration laboratories)

- ☞ Having well practical knowledge of Micro& Macro structure analysis and Failure Analysis by using Envision6.0 image analyzer
- ☞ Handling Durability test by using Servo Hydraulic Machine with MTS Multipurpose elite test.
- ☞ Responsible for raw material Inspection Coating Thickness measurement and Weld Penetration
- ☞ Responsible for IGC , Sulfide stress corrosion cracking , HIC and Salt spray testing
- ☞ Responsible for Cyclic corrosion test by using ASCOTT 2000ip UK make Instrument (Cyclic corrosion test chamber)
- ☞ Customer evaluation and maintaining quality parameters for incoming steel coils, Cast iron and welded Samples
- ☞ Support to the all customer related complaints and improvements

May'16-may'17 with LGB Pvt Ltd, Coimbatore as Trainee & Junior Engineer- Heat treatment & Metallurgical Lab

Role:

- ☞ Responsible for through hardening and case hardening of low carbon and Medium carbon steel & quenching followed by suitable tempering process by use rotary tilting furnace.
- ☞ Handling problem solving, cost reduction and rejection control failure analysis for the products.
- ☞ To inspect the Metallurgical parameters of incoming raw materials for chain components.
- ☞ Responsible for daily production plan.
- ☞ Responsible for having periodical check and follow up activities in order to technical problems if any incurred during routine production.
- ☞ To ensure the Metallurgical parameters for heat treated chain.
- ☞ Responsible for reporting to the top management regarding daily production.
- ☞ Providing ongoing problem solving/ continuous support to Manufacturing for rejection reduction & Re-work Cost reduction.

TRAININGS/ EVENTS ATTENDED

- ☞ **Venue:** SAIL , Salem Steel Plant – Salem
- ☞ **Title :** Microstructure Observation of Various Grade of Stainless steel Coils
- ☞ **Duration:** 2 month
- ☞ **Team size:** 4
- ☞ **Short description:** Analysis the microstructure of AISI 304, 304L, 316L, Duplex and Super duplex Stainless Steels microstructure studied

PERSONAL DETAILS

- ☞ **Address** : 5/215, Thimmasi valavu, Periya soragai (Po),Salem(Dt), Tamilnadu-636502.
- ☞ **Date of Birth** : 05-01-1993
- ☞ **Languages known** : English, Tamil,
- ☞ **Passport No.** : N8602091
- ☞ **Hobby** : Playing and watching Cricket, Listening to music

I hereby declare that all the information given above is true to the best of my knowledge.

Date:

(SURESH PALANIVEL)